LEARNING MODULE

25

Time Series I: ARMA, Unit Roots and Cointegration

A price index wanders with no level to return to, while the futures–spot basis keeps coming back. Stationarity, ARMA models, unit-root tests and cointegration say which is which, and how fast.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM6 · LM12 · LM16 · LM17 · LM18 · LM19 · LM20 · LM21 · LM23

LEADS TO LM26 · LM33 · LM34 · LM36 · LM39 **TIME** ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 define weak stationarity, white noise and a random walk, and explain why a price level is not stationary while its changes can be
☐	2 derive the autocorrelations and partial autocorrelations of AR(1) and MA(1) models, estimate an AR(1) by least squares and convert its coefficient into a half-life
☐	3 test a return series for autocorrelation with the iid and the robust bands, the Ljung–Box test and the variance-ratio test, and explain how volatility clustering changes the verdict
☐	4 explain why the Dickey–Fuller t-stat does not follow a t distribution, and apply the augmented test with MacKinnon p-values and the KPSS test of the reverse null
☐	5 explain why regressions between independent random walks look significant, and test cointegration with a known vector and with an estimated one
☐	6 decompose the TX–TAIEX log basis into convergence at expiry, roll jumps, the dividend season and one-day noise, and measure how fast it reverts
☐	7 run these tools in Python with statsmodels on NumPy arrays, matching each package result to a calculation by hand

LEARNING MODULE OVERVIEW

- A random walk's variance grows like t, so a price level has no mean to return to; its changes can be stationary. TAIEX rose 10.9-fold in 2003–2026, and its log shows no evidence against a unit root (ADF −0.77, p-value 0.97).
- An AR(1) has autocorrelations ϕ_1^k and a half-life $\ln \frac{1}{2} / \ln \phi_1$, (25.1)–(25.2); an MA(1) remembers one day, (25.3). The TX–TAIEX log basis has an AR(1) coefficient of 0.7758, a half-life of 2.73 sessions.
- Volatility clustering widens the band of a sample autocorrelation, (25.4). TAIEX's lag-1 value 0.0393 has a t-stat of 3.01 against the iid band and 1.92 against the robust one; Ljung–Box (25.5) and the variance-ratio test (25.6) need the same care.
- Under a unit root without drift the Dickey–Fuller t-stat (25.7) is shifted left: its 5% point is −2.86, and a true random walk falls below −1.645 in 46.7% of simulated samples. The basis rejects a unit root (ADF −9.00).
- Two independent random walks look related: their regression has a t-stat beyond ±1.96 in 96.6% of 1,000 simulated pairs. The log basis is a cointegrating combination with the known vector (1, −1).
- The basis has slow parts: convergence at expiry, roll jumps (−144.7 bp at the June roll) and the dividend season, with June's mean −74.80 bp of which −67.43 bp is carry and dividends.
- Read as a persistent AR(1) plus one-day noise, (25.9), the basis's ARMA(1,1) fit puts 64% of the variance of a day's change in noise; the persistent part halves in 7.04 sessions (SE 0.556).

1 INTRODUCTION

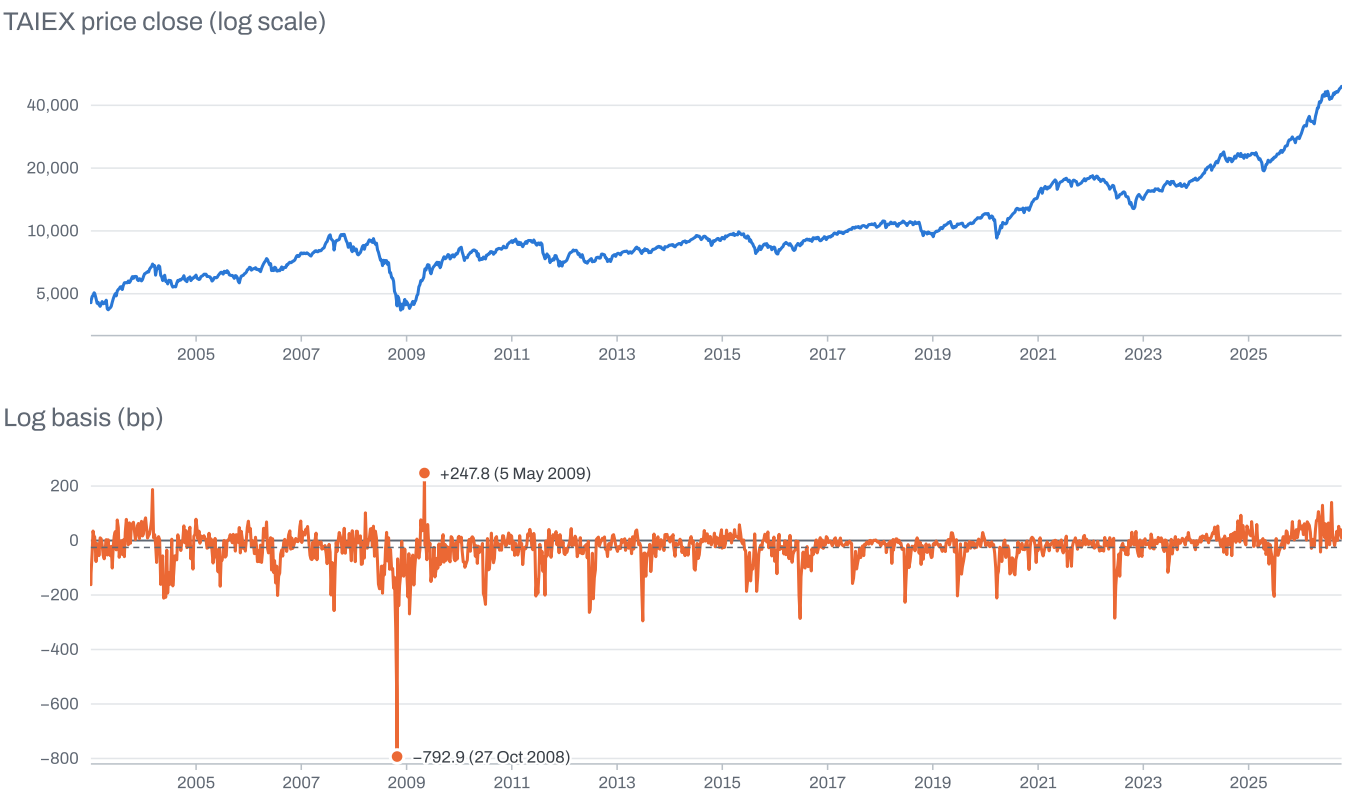
A price index has no level it must return to, while a futures price is tied to its index by carry and dividends. This module turns that difference into models and tests, applied to the TX near-month future and the TAIEX.

MOTIVATING CASE **A basis that comes back**

From 2 January 2003 to 8 October 2026 the TAIEX price index rose from 4,524.87 to 49,313.44, 10.9 times its start, and its log never settled around a level. The near-month TX future followed it. Their log basis, $b_t = \ln F_t - \ln S_t$, averaged −25.4 bp with a standard deviation of 58.2 bp (Exhibit 1). It reached −793 bp on 27 October 2008, when TX settled 7.0% lower and the index 4.6% lower, and +248 bp on 5 May 2009.

The future settled below the index on 4,051 of 5,850 days. A day's change in the basis had a standard deviation of 38.9 bp, yet the basis did not wander off. Next-day values correlate 0.776, so an AR(1) would halve a gap in about 2.7 sessions. LM6 split one season's basis into carry, dividends and a residual. Is the basis stationary while the index is not? How fast does it return, and how much of a day's move is noise?

Exhibit 1: The TAIEX and the TX–TAIEX log basis, 2003–2026



Note: top, S_t (no dividends), log scale; bottom, b_t in bp, its mean −25.4 dashed and the extremes −792.9 and +247.8 marked. TX23 at weekly points: each ISO week's last session, the first date and the extremes, 1,227 of 5,850 dates. Descriptive. Source: FinMind `TaiwanFuturesDaily` (TX, regular session), retrieved 19 October 2026, and `TaiwanStockPrice` (TAIEX), retrieved 11 and 18 October 2026. Computations: `data/lm25_numbers.py` and `data/lm25_charts.py`.

Sections 2 to 6 build the tools, from stationarity to cointegration. Section 7 resolves the case, and Section 8 runs the tools in Python.

> **NOTE** **Notation for this module**
>
> TX23 holds 5,850 dates. On each, F_t is the settlement of the TX near-month contract held from that close, and S_t the TAIEX price close. The contract rolls at the close of the day before its last trading day. The log basis $b_t := \ln F_t - \ln S_t$ is in basis points (bp, 10^{-4}); LM6's $F - S$ was in points. A rolled change differences the stored series; a roll-free change uses one contract on both days.
>
> D holds the 5,834 dates, to 14 September 2026, whose contract expires within the sample (the last, 202609, on 16 September). TR23's $T = 5{,}849$ log returns r_t serve the autocorrelation tests.
>
> Symbols with the book's usual meaning: γ_k, ρ_k, $\hat{\rho}_k$, η_t, σ_η^2, ϕ_1, θ_1, δ, β, ε_t; c an AR intercept; τ, q, $r_f = 2\%$ as in LM6; m the variance-ratio horizon; ℓ a log-likelihood; x_t, y_t generic series. Local, displacing book-wide letters this module does not use: b_t; a_j, J the ADF test's lag coefficients and their number; u_t, σ_u one-day noise; ζ_t, σ_ζ the persistent part's innovation; $Q_{\mathrm{LB}}(K)$ over K lags; $\mathrm{SE}_{\mathrm{iid}}$, $\mathrm{SE}_{\mathrm{rob}}$; $\tilde{r}_t := r_t - \bar{r}$; α_F, α_S; V_t a Brownian bridge. Standard deviations divide by the number of values.

NOTE **Research hygiene**

Basis figures use TX23, or D where a caption says so. The residual of (6.25) uses dividends that went ex up to each contract's expiry: a look-ahead that suits a decomposition, never a trading rule.

Tests and fits are **in-sample**, the limit regimes a **stability check**, and the simulations and widget a **model**. The figures of the case, Example 1 and the expiry, season and roll figures are **descriptive**, without standard errors: they describe the sample, and neighbouring days correlate about 0.78. Nothing is tested on later data (LM34).

Tests counted (LM21's rule): 36 on TR23's log returns, one of them Problem 6's $Q_{\mathrm{LB}}(5)$, with 1.8 false rejections expected at the 5% level. Two are on $|r_t|$ and 24 on TX23. Every DF and ADF verdict lies far from its critical value.

Bonferroni's correction for 24 tests needs a KPSS statistic above 1.03, the 0.05/24 upper point of its limit law (Kwiatkowski et al. 1992). That law is the law of $\int_0^1 V_t^2\,dt$ for a Brownian bridge V_t (LM36), the Cramér–von Mises limit law (Csörgő and Faraway 1996, p. 222), which SciPy evaluates.

The five KPSS statistics for the basis, 0.62 to 0.96, fall short, and so does $\hat{\alpha}_F$'s t-stat, −2.9, against the two-sided normal point 3.08. The yearly means also show a moving mean (Example 6).

2 STATIONARITY, WHITE NOISE AND THE RANDOM WALK

- [ ] define weak stationarity, white noise and a random walk, and explain why a price level is not stationary while its changes can be

A **time series** is a sequence $x_1, \dots, x_T$ observed at equal steps, here trading sessions, and treated as one draw of random variables x_t. Its **autocovariance** at lag k is $\mathrm{Cov}(x_t, x_{t-k})$, (16.2).

DEFINITION **Weak stationarity**

A series x_t is **weakly stationary** when

1. its mean $\mathbb{E}x_t$ is the same for every t;
2. its variance $\gamma_0 := \mathrm{Var}\, x_t < \infty$ is the same for every t;
3. its autocovariance $\gamma_k := \mathrm{Cov}(x_t, x_{t-k})$ depends only on the lag k.

The series' **autocorrelations** $\rho_k := \gamma_k/\gamma_0$ then depend only on k as well.

DEFINITION **White noise**

A series η_t is **white noise** with variance σ_η^2 when $\mathbb{E}\eta_t = 0$, $\operatorname{Var}\eta_t = \sigma_\eta^2 < \infty$ and $\operatorname{Cov}(\eta_t, \eta_{t-k}) = 0$ for every lag $k \geq 1$.

Independent draws with mean 0 and a common finite variance are white noise by (16.1). The converse fails, because white noise only needs zero correlation. TAIEX's daily log returns are close to uncorrelated, but their sizes are not: the lag-1 autocorrelation of $|r_t|$ is 0.161 (LM19). Volatility that changes over time is modelled in LM26; white noise here only needs a constant variance.

A **random walk** adds white noise to its last value, $x_t = x_{t-1} + \eta_t$; with a **drift** c it is $x_t = c + x_{t-1} + \eta_t$. After t steps $x_t - x_0 = ct + \eta_1 + \cdots + \eta_t$, and the terms are uncorrelated, so (16.3) gives

$$\operatorname{Var}(x_t - x_0) = t\,\sigma_\eta^2.$$

This variance grows without bound, so a random walk is not stationary, while its changes $\Delta x_t = c + \eta_t$ are. LM3 treated $\ln S$ as such a walk.

EXAMPLE 1

A level and a spread across two regimes

TWSE's daily price limit widened from 7% to 10% on 1 June 2015, which splits TX23 into 3,078 and 2,772 dates. Compare the mean of S_t and the mean and standard deviation of b_t in the two regimes. Then give the standard deviation of a random walk after t sessions.

SOLUTION

The index averaged 7,302 before the change and 16,042 after it: a level's mean depends on the window. The basis averaged −28.6 bp and −21.8 bp, with standard deviations of 65.7 and 48.2 bp, so the spread moved much less. A random walk's standard deviation after t sessions is $\sigma_\eta\sqrt{t}$, so it has no fixed variance.

PITFALL **A sample mean of a level is not a mean to return to**

The mean TAIEX close over 2003–2026 describes the path the index happened to take. A random walk has no mean to return to, so a level "below its long-run average" is no forecast. Only a stationary series has a mean that pulls it back.

導讀 **價格水準沒有回得去的平均，變動才可能有**

時間序列（time series）是依時間排列的一串觀測值。平穩（weakly stationary）要求平均數（mean）與變異數（variance）固定，兩期之間的共變異數（covariance）只跟間隔有關。白噪音（white noise）只要求平均為零、變異數固定、互不相關（uncorrelated）；平均為零、變異數有限的獨立同分布（iid）序列一定是白噪音，反之則不然，例如報酬幾乎互不相關，大小卻會成群出現。隨機漫步（random walk）把白噪音一期一期加上去，t 期後的變異數是 $t\sigma_\eta^2$，越走越散，所以價格指數沒有一個會拉它回去的平均。它的變動才是平穩的，這就是分析報酬而不分析價格水準的原因。

KNOWLEDGE CHECK 1

A random walk has a daily standard deviation of 1.2%. What is the standard deviation of its change over 25 and over 100 sessions? Compare $x_t = 0.9x_{t-1} + \eta_t$ with the same σ_η, whose variance settles at $\sigma_\eta^2/(1-0.9^2)$ (Section 3 derives it).

ANSWER

By $\sigma_\eta\sqrt{t}$, 6.0% after 25 sessions and 12.0% after 100, growing without bound. The second series' standard deviation settles at $1.2\%/\sqrt{0.19} = 2.753\%$, however long the horizon.

3 AUTOREGRESSIVE AND MOVING-AVERAGE MODELS

- [ ] derive the autocorrelations and partial autocorrelations of AR(1) and MA(1) models, estimate an AR(1) by least squares and convert its coefficient into a half-life

The basis remembers yesterday: next-day values correlate 0.776. The simplest model with such memory regresses today's value on yesterday's.

RESULT **The AR(1) model**

Let η_t be white noise with variance σ_η^2, uncorrelated with $x_{t-1}, x_{t-2}, \ldots$, and let

$$x_t = c + \phi_1 x_{t-1} + \eta_t. \tag{25.1}$$

If x_t is weakly stationary, then $|\phi_1| < 1$, $\mathbb{E}x_t = c/(1-\phi_1)$, $\operatorname{Var} x_t = \sigma_\eta^2/(1-\phi_1^2)$ and $\rho_k = \phi_1^k$. At $\phi_1 = 1$ the model is a random walk with drift c.

Take expectations of (25.1): $\mathbb{E}x_t = c + \phi_1 \mathbb{E}x_t$. Take variances: $\gamma_0 = \phi_1^2 \gamma_0 + \sigma_\eta^2$, because η_t is uncorrelated with x_{t-1}, and a positive γ_0 needs $\phi_1^2 < 1$. Multiplying the centred equation by x_{t-k} and taking expectations gives $\gamma_k = \phi_1 \gamma_{k-1}$ (Problem 7).

When the η_t are independent of the past, today's gap from the mean is expected to shrink by the factor ϕ_1 each session. For $0 < \phi_1 < 1$ the gap halves after

$$\text{half-life} = \frac{\ln \frac{1}{2}}{\ln \phi_1} \text{ sessions.} \tag{25.2}$$

A **moving-average** model remembers shocks rather than values. The MA(1) is $x_t = c + \eta_t + \theta_1 \eta_{t-1}$, stationary for every θ_1. Its variance is $(1 + \theta_1^2)\sigma_\eta^2$ and $\operatorname{Cov}(x_t, x_{t-1}) = \theta_1 \sigma_\eta^2$ by (16.3) and (16.4), while values two or more sessions apart share no η. So

$$\rho_1 = \frac{\theta_1}{1 + \theta_1^2}, \qquad \rho_k = 0 \quad (k \geq 2). \tag{25.3}$$

The **partial autocorrelation** at lag k is the last coefficient of an OLS regression of x_t on a constant and $x_{t-1}, \ldots, x_{t-k}$. It measures the link with x_{t-k} that the nearer lags leave unexplained.

An AR(1) has autocorrelations that decay and partial autocorrelations that stop after lag 1: given x_{t-1}, earlier lags add nothing, because η_t is uncorrelated with all of them. An MA(1) is the reverse: with $\rho_2 = 0$, the normal equations of the two-lag regression give a lag-2 coefficient of $-\rho_1^2/(1 - \rho_1^2) \neq 0$.

The **ARMA(1,1)** model combines the two, $x_t = c + \phi_1 x_{t-1} + \eta_t + \theta_1 \eta_{t-1}$ with $|\phi_1| < 1$. Its autocorrelations fall by the factor ϕ_1 per lag from

$$\rho_1 = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)}{1 + 2\phi_1\theta_1 + \theta_1^2}$$

(stated without proof; Hamilton 1994, ch. 3). OLS and its classical standard error are LM23's, (23.1)–(23.3); the AR(1) estimate is (12.1) with a constant and one lagged value.

EXAMPLE 2

An AR(1) for the basis

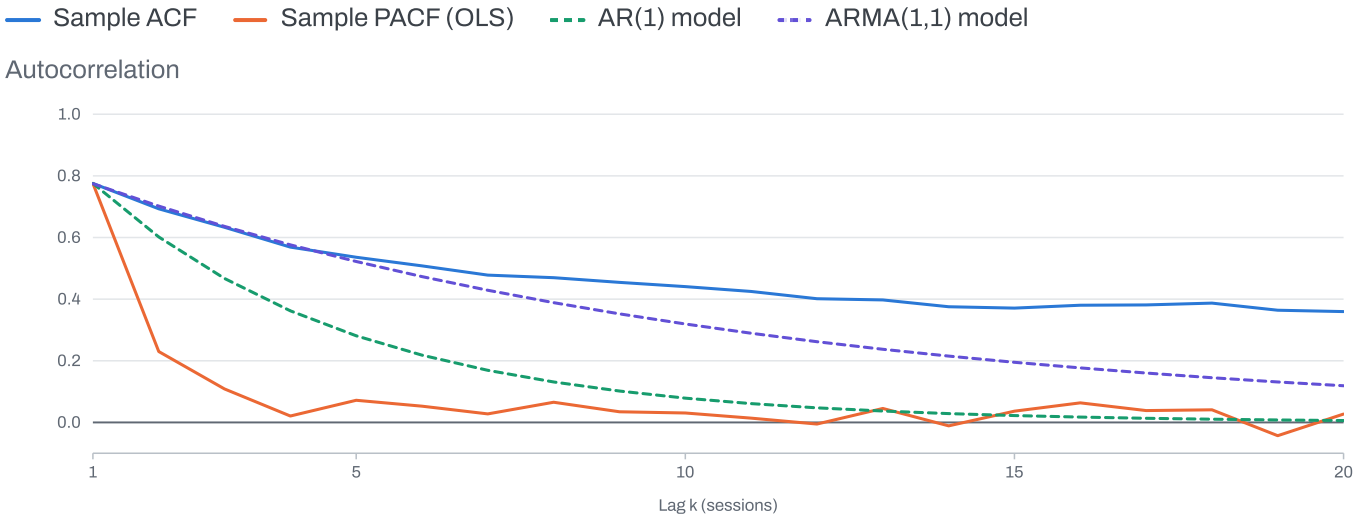
Regress b_t (bp) on a constant and b_{t-1} by OLS over TX23's 5,849 rolled changes. Report $\hat{c}$, $\hat{\phi}_1$ with its classical standard error, s_ε, the half-life and the implied mean.

SOLUTION

OLS gives $\hat{c} = -5.66$ bp, $\hat{\phi}_1 = 0.7758$ (classical SE 0.00824) and $s_\varepsilon = 36.7$ bp. By (25.2) the half-life is $\ln \frac{1}{2} / \ln 0.7758 = 2.73$ sessions, and the implied mean is $\hat{c}/(1 - \hat{\phi}_1) = -25.25$ bp. The classical standard error assumes white-noise errors, which the partial autocorrelation of 0.230 at lag 2 denies. So the half-life gets no interval here; Example 10 gives one. Roll-free changes give 0.7748 and 2.72 sessions.

Exhibit 2 compares the basis with two models. Its partial autocorrelations of 0.230 and 0.109 at lags 2 and 3 exclude a pure AR(1). Beyond lag 5 the sample autocorrelations decay more slowly than either model, because the mean itself moves (Section 7).

Exhibit 2: Autocorrelations and partial autocorrelations of the log basis



Note: TX23, lags 1 to 20. Sample autocorrelations (19.5): 0.776, 0.693, 0.633 at lags 1 to 3, 0.441 at 10, 0.360 at 20. Partial autocorrelations by OLS: 0.776, 0.230, 0.109, 0.021, 0.072, then below 0.07. Models: AR(1) 0.7758^k (Example 2), 0.602 at lag 2, 0.281 at 5, 0.079 at 10; ARMA(1,1) (Example 10), 0.7745, 0.702, 0.522, 0.319 and 0.119 at lags 1, 2, 5, 10 and 20. In-sample. Source: as Exhibit 1.

PITFALL **A half-life from a noisy series is too short**

One-day noise on top of a persistent series lowers $\hat{\rho}_1$, and with it the AR(1) half-life. The basis's AR(1) gives 2.7 sessions; Section 7 separates the noise and finds 7.0 sessions for the persistent part.

導讀 **自我迴歸與移動平均：記住昨天，或記住昨天的衝擊項**

自我迴歸模型（autoregressive model, AR）把今天的值寫成昨天的值乘上 ϕ_1 再加白噪音（white noise），所以自我相關函數（autocorrelation function, ACF）是 ϕ_1^k，一期期衰減。偏離平均數（mean）的缺口每期縮成 ϕ_1 倍，縮到一半所需的期數就是半衰期（half-life）$\ln\frac{1}{2}/\ln\phi_1$。移動平均模型（moving average model, MA）記住的是昨天的衝擊項（shock），自我相關只到第一期就歸零。偏自我相關函數（partial autocorrelation function, PACF）扣掉較近各期的影響後再看第 k 期，AR(1) 的 PACF 在第一期之後歸零，MA(1) 則相反。自我迴歸移動平均模型（ARMA）結合兩者，第 7 節會用它把基差（basis）拆成持續的部分與一日雜訊（noise）。

KNOWLEDGE CHECK 2

An MA(1) has $\theta_1 = -0.36$. Find ρ_1 and ρ_2. What is the half-life of an AR(1) with $\phi_1 = 0.5$?

ANSWER

By (25.3), $\rho_1 = -0.36/1.1296 = -0.319$ and $\rho_2 = 0$. By (25.2), $\ln \tfrac{1}{2} / \ln 0.5 = 1$ session.

4 TESTING FOR AUTOCORRELATION

- ☐ test a return series for autocorrelation with the iid and the robust bands, the Ljung–Box test and the variance-ratio test, and explain how volatility clustering changes the verdict

Is TAIEX's lag-1 autocorrelation of log returns, 0.0393 (LM19), evidence of memory? LM19's iid band $\pm 1.96/\sqrt{T}$, (19.6), rests on a normal approximation that LM19 states without proof, beyond LM17's (17.5). Tests and p-values are LM21's (21.2), and each p-value comes with the number of tests run on the same data.

The iid band assumes independent returns. Clustered returns pair large moves with large moves, so the products $\tilde{r}_t \tilde{r}_{t-k}$ vary more than under independence, and $\hat{\rho}_k$ with them.

RESULT **A robust standard error of $\hat{\rho}_k$**

Let the returns r_t have conditional mean $\mathbb{E}r$ given the past (a null hypothesis that implies no autocorrelation) and finite fourth moments, and let $\tilde{r}_t := r_t - \bar{r}$. Then, for large T,

$$\mathrm{SE}_{\mathrm{rob}}(\hat{\rho}_k) = \frac{\sqrt{\sum_{t>k} \tilde{r}_t^2 \tilde{r}_{t-k}^2}}{\sum_t \tilde{r}_t^2}. \tag{25.4}$$

For iid returns it is close to $\mathrm{SE}_{\mathrm{iid}} = 1/\sqrt{T}$.

Formula (25.4) is HC0-type, the lag-k version of LM23's White errors (23.6); the Deep Dive derives it.

EXAMPLE 3

Is the lag-1 autocorrelation evidence?

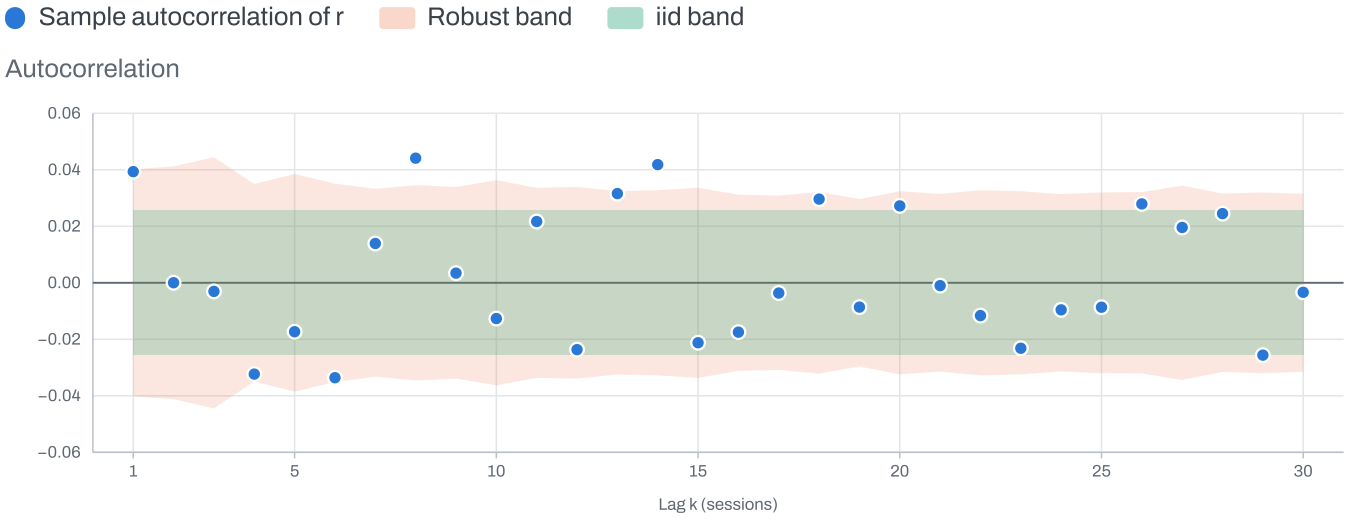
TR23's log returns have $\hat{\rho}_1 = 0.0393$ with $T = 5{,}849$. Test $\rho_1 = 0$ at the 5% level (two-sided, normal reference) with $\mathrm{SE}_{\mathrm{iid}}$ and with (25.4), whose value at lag 1 is 0.0205.

SOLUTION

The iid standard error is $1/\sqrt{5849} = 0.013076$, so the t-stat is $0.0393/0.013076 = 3.01$, and the test rejects. The robust t-stat is $0.0393/0.0205 = 1.92$, below 1.96: once clustering is allowed for, the lag-1 correlation is no evidence at the 5% level. Over lags 1 to 30, 9 values leave the iid band and 2 leave the robust one, against $0.05 \times 30 = 1.5$ expected under the null (Exhibit 3).

Formula (25.4) needs a finite fourth moment, which TR23 may lack (LM19), so its t-stats are approximate. Leaving the product of 9 and 10 April 2025 out of (25.4)'s sum lowers the standard error to 0.01954 and lifts the t-stat to 2.01.

Exhibit 3: TAIEX log-return autocorrelations against the iid and robust bands



Note: TR23 log returns, $T = 5{,}849$, lags 1 to 30; $\hat{\rho}_k$ by (19.5). Bands of ± 1.96 standard errors: iid, ± 0.0256; robust, by (25.4), with standard errors from 0.0151 to 0.0227. Outside the iid band: lags 1, 4, 6, 8, 13, 14, 18, 20 and 26 (lag 29 lies inside by 2×10^{-5}). Outside the robust band: lags 8 and 14. In-sample; 36 tests on these returns, one in Problem 6. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

The **Ljung–Box test** combines lags 1 to K in one statistic,

$$Q_{\mathrm{LB}}(K) = T(T+2) \sum_{k=1}^{K} \frac{\hat{\rho}_k^2}{T-k}. \tag{25.5}$$

For iid data with a finite variance and large T it is approximately χ^2_K (stated without proof; Ljung and Box 1978), and large values reject.

LM19's identity column shows $\mathrm{VR}(m) - 1$ close to $2\sum_{k<m}(1-k/m)\hat{\rho}_k$. If the $\hat{\rho}_k$ are uncorrelated, that sum has the variance $\mathrm{SE}^2_{\mathrm{rob}}$ of (25.6), and the **variance-ratio test** is

$$\begin{aligned} \text{t-stat} &= \frac{\mathrm{VR}(m)-1}{\mathrm{SE}}, \\ \mathrm{SE}^2_{\text{iid}} &= \frac{2(2m-1)(m-1)}{3mT}, \\ \mathrm{SE}^2_{\text{rob}} &= \sum_{k=1}^{m-1}\left[2\left(1-\frac{k}{m}\right)\right]^2 \mathrm{SE}_{\text{rob}}(\hat\rho_k)^2. \end{aligned} \tag{25.6}$$

The iid form puts $1/T$ for each $\mathrm{SE}_{\text{rob}}(\hat\rho_k)^2$. The test is two-sided with a normal reference, a large-sample approximation; its robust form also assumes that the Deep Dive's cross-lag terms vanish.

EXAMPLE 4

Ljung–Box and variance ratios for TAIEX

On TR23's log returns, compute $Q_{\text{LB}}(10)$ and the t-stats (25.6) at m = 2, 5, 10 and 20, each at the 5% level.

SOLUTION

The statistic $Q_{\text{LB}}(10)$ is 37.1 against the χ^2_{10} 5% point 18.31, a p-value of 5.4×10^{-5}. Its χ^2 law needs iid data, so clustering may cause the rejection (Problem 6).

LM19's variance ratios, with iid and robust t-stats, are 1.0392 (3.00, 1.91) at $m=2$, 1.0465 (1.62, 1.02) at 5, 1.0065 (0.15, 0.09) at 10 and 1.0359 (0.55, 0.37) at 20. Only the iid test at $m=2$ rejects. Keeping the cross-lag terms widens the robust standard error by up to 25% and changes no verdict (t-stats 0.90, 0.08 and 0.30 at m = 5, 10 and 20). For $|r_t|$, $Q_{\text{LB}}(10)$ is 1,820: sizes cluster (LM26).

DEEP DIVE **Where the robust band comes from** optional · click to expand

Here $e_t := r_t - \mathbb{E}r$ is a return's deviation from its true mean. Under the null each e_t has conditional mean 0 given the past. The product $e_t e_{t-k}$ then has conditional mean 0 given days up to $t-1$, which fix every earlier product, so the products are uncorrelated. By (16.3),

$$\mathrm{Var}\left(\sum_{t>k} e_t e_{t-k}\right) = \sum_{t>k}\mathbb{E}\left[e_t^2 e_{t-k}^2\right].$$

Dividing by $(\sum_t e_t^2)^2$, then replacing $\mathbb{E}r$ by $\bar r$ and the expectations by sample values, a large-T approximation, gives (25.4). Formula (25.4) is (23.6)'s White form for a regression of $\tilde r_t$ on $\tilde r_{t-k}$, evaluated under the null. For iid returns each expectation is $(\mathrm{Var}\,r)^2$, and (25.4) is close to $1/\sqrt{T}$. For two lags $j\neq k$ the argument leaves cross terms $\mathbb{E}[e_t^2 e_{t-j}e_{t-k}]$, which the robust form of (25.6) sets to zero.

PITFALL **An iid band on clustered data**

The iid band is too narrow for clustered returns, and Ljung–Box's χ^2_K law has the same flaw. Use (25.4).

導讀 **波動群聚讓獨立同分布區間太窄**

自我相關（autocorrelation）的獨立同分布區間（iid band）$\pm 1.96/\sqrt{T}$ 是在獨立（independent）報酬的虛無假設（null hypothesis）下算出來的。可是加權指數有波動群聚現象（volatility clustering）：大漲大跌成群出現，相乘項 $\tilde{r}_t\tilde{r}_{t-k}$ 的波動比獨立時大，樣本自我相關也跟著跳得更大。穩健標準誤（robust standard error）把這些相乘項的平方直接加總，落後一期的標準誤從 0.0131 變成 0.0205，t 統計量（t-statistic）從 3.01 降到 1.92，不再顯著（significant）。Ljung–Box 檢定（Ljung–Box test）與用 iid 標準誤的變異數比（variance ratio）檢定也假設獨立，同樣會誤判。報酬幾乎互不相關（uncorrelated），但大小彼此相關（correlated），這是 LM26 的主題。

KNOWLEDGE CHECK 3

LM19 found a lag-1 autocorrelation of 0.161 for $|r_t|$, and (25.4) gives it a robust standard error of 0.0240. Is it significant at the 5% level?

ANSWER

Yes. The robust t-stat is $0.161/0.0240 = 6.7$, and 0.161 lies far outside the iid band ± 0.0256 too. Clustering is dependence of sizes, which LM26 models.

5 UNIT ROOTS AND THE DICKEY–FULLER TEST

- ☐ explain why the Dickey–Fuller t-stat does not follow a t distribution, and apply the augmented test with MacKinnon p-values and the KPSS test of the reverse null

Is $\ln S_t$ a random walk, or an AR(1) with ϕ_1 just below 1? Subtract x_{t-1} from both sides of (25.1):

$$\Delta x_t = c + \delta x_{t-1} + \eta_t, \qquad \delta := \phi_1 - 1. \tag{25.7}$$

A **unit root**, $\phi_1 = 1$, is $\delta = 0$. The **Dickey–Fuller test** (Dickey and Fuller 1979) tests $H_0 : \delta = 0$ against $H_1 : \delta < 0$, one-sided. Its statistic, the DF t-stat, is $\hat{\delta}/\operatorname{SE}(\hat{\delta})$ from OLS with the classical standard error,

and small values reject. Under H_0 with no drift ($c = 0$ too) it follows neither the normal nor the t law of (23.4): its limit is a functional of Brownian motion (LM36), simulated here.

With a drift, the statistic of a regression with a constant only is asymptotically normal instead (stated without proof; Hamilton 1994, §17.4, Case 3). With a trend in the regression, it follows the trend case's own Dickey–Fuller law, not the one simulated here, whatever the drift (Case 4, also stated without proof). So a drifting series is tested with a trend (below).

EXAMPLE 5

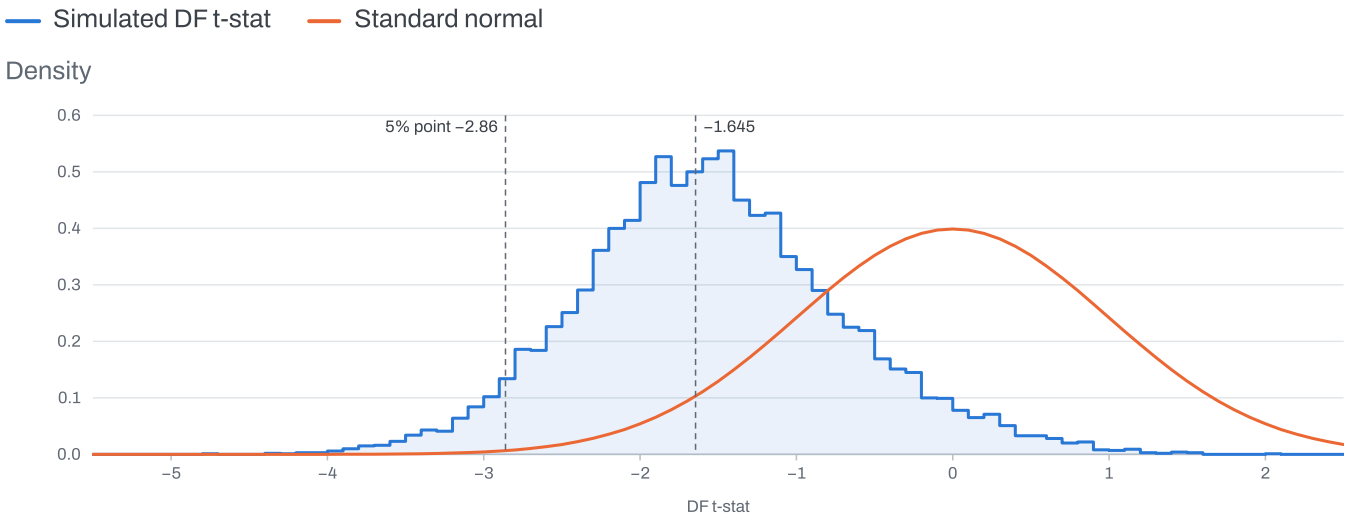
The Dickey–Fuller null by simulation

Experiment 1 (*Model*, seed `[1, 20261011]`) draws 10,000 random walks $x_t = z_1 + \cdots + z_t$ of 5,850 levels, with z_t iid standard normal. Each walk gives a DF t-stat with a constant and no lags. Find the 1%, 5% and 10% points and the share below −1.645.

SOLUTION

The points are −3.44, −2.86 (batch SE 0.014, LM18) and −2.58, and the mean is −1.54 (Exhibit 4). For 5,849 observations, statsmodels gives MacKinnon's (2010) points as −3.43, −2.86 and −2.57. A true unit root falls below the normal 5% point −1.645 in 46.7% of the draws (SE 0.5%), and below −1.96 in 30.7%.

Exhibit 4: The DF t-stat under a unit root against the standard normal



Note: histogram, in bins of 0.1, of 10,000 DF t-stats with a constant and no lags, from random walks of 5,850 levels (experiment 1), and the standard normal density. Dashed: the simulated 5% point −2.86, equal to MacKinnon's to two decimals, and the normal −1.645. *Model.*

When the changes are autocorrelated, the errors of (25.7) are not white noise. The **augmented Dickey–Fuller** (ADF) test adds J lagged changes, which approximate an ARMA by an autoregression (Said and Dickey 1984):

$$\Delta x_t = c + \delta x_{t-1} + \sum_{j=1}^{J} a_j \, \Delta x_{t-j} + \eta_t. \tag{25.8}$$

statsmodels' `adfuller` picks J by the Akaike information criterion (AIC), which trades fit against lags, from 0 to $\lceil 12(T/100)^{1/4} \rceil$, 34 here. It reports MacKinnon's p-values (MacKinnon 1994, 2010). A drifting series also gets a trend term, `regression="ct"`, because its alternative is stationarity around a trend.

The **KPSS test** (Kwiatkowski et al. 1992) reverses the null: H_0 is stationarity around a constant, and large values reject (one-sided). Its statistic grows with the squared partial sums of the demeaned series, which wander under a unit root. statsmodels interpolates its p-value in a table of four points (0.347, 0.463, 0.574 and 0.739 at 10%, 5%, 2.5% and 1%), so p-values stop at 0.01 and 0.10.

EXAMPLE 6

Unit-root tests on the index and on the basis

Test $\ln S_t$ and b_t on TX23 at the 5% level: the DF regression by hand, then ADF, then KPSS.

SOLUTION

For $\ln S_t$ the DF t-stat with a constant and no lags is 1.19, with $\hat{\phi}_1 = 1.000392$. The no-drift p-value, 0.996, does not apply to a drifting index, but a positive statistic rejects a unit root under neither law. ADF with a trend (32 lags by AIC) gives −0.77, p-value 0.97: no evidence against a unit root. KPSS gives 10.356, p-value below 0.01.

For b_t the DF t-stat is −27.20, and ADF with a constant (28 lags) gives −9.00, p-value below 10^{-14}. KPSS gives 0.664, between the 2.5% and 1% points, p-value 0.017. Both tests reject for the basis: it returns within days to a mean that moves across years, from −80 bp in 2008 to +34 bp in 2026 to date.

DEEP DIVE **Why the DF t-stat shifts left** optional · click to expand

Under $\delta = 0$ and $c = 0$ the regressor x_{t-1} is a random walk, so $\mathbb{E} \sum_t x_{t-1}^2$ grows like T^2, not like T. The estimate $\hat{\delta}$ settles at the rate T, and the t-stat converges to a ratio of functionals of a standard Brownian motion B_t (LM36), not to $\mathcal{N}(0,1)$. With a constant the limit is negative on average: experiment 1's mean is −1.54. Stated without proof (Hamilton 1994, ch. 17; MacKinnon 1996 for the tables).

PITFALL **Reading the DF t-stat against −1.645**

Against −1.645, a test at the nominal 5% level rejects a true unit root in 46.7% of samples. For a series without drift, use MacKinnon's point, −2.86 with a constant and no trend, or the p-value of `adfuller`.

導讀 **單根檢定：別拿 −1.645 當標準**

單根（unit root）就是 $\phi_1 = 1$，序列成了隨機漫步（random walk）。單根檢定（unit root test）的虛無假設（null hypothesis）是 $\delta = 0$，對立假設（alternative hypothesis）是 $\delta < 0$，屬於單邊檢定（one-sided test）。問題在於虛無假設成立、序列又沒有漂移項（drift）時，水準的平方和以 T^2 的速度成長，t 統計量（t-statistic）不再近似常態分布（normal distribution），而是整體往左移：模擬一萬條隨機漫步，有 46.7% 落在 −1.645 以下。所以要用 MacKinnon 的臨界值（critical value），5% 約為 −2.86。增廣 Dickey-Fuller 檢定（augmented Dickey–Fuller test）加入前幾期的變動（lagged changes），吸收殘差（residual）的自我相關（autocorrelation）；KPSS 檢定（KPSS test）則反過來，以平穩性（stationarity）為虛無假設。

KNOWLEDGE CHECK 4

Adding 28 lagged changes moves the basis's statistic from −27.20 to −9.00. Why does it move, and why does the verdict stay?

ANSWER

The lagged changes absorb the part of the reversion that the partial autocorrelations show at lags 2 and 3 (0.230 and 0.109), so $\hat{\delta}$ alone carries less of it. Both statistics lie far below MacKinnon's 1% point, −3.43.

6 SPURIOUS REGRESSION AND COINTEGRATION

- [ ] explain why regressions between independent random walks look significant, and test cointegration with a known vector and with an estimated one

Regress one random walk on another, independent one: OLS finds a link almost every time.

EXAMPLE 7

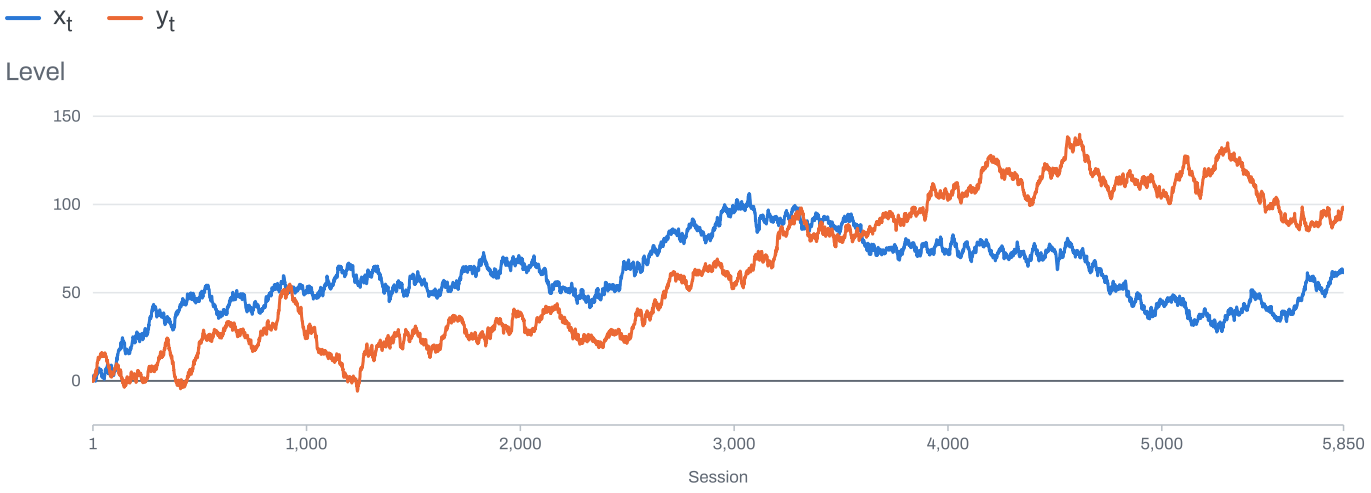
Two unrelated random walks

Experiment 2 (*Model*, seed `[2, 20261011]`) draws 1,000 independent pairs of random walks of 5,850 sessions, x_t then y_t. Regress y_t on a constant and x_t by OLS with the classical standard error. How often is $|\text{t-stat}| > 1.96$?

SOLUTION

The first pair (Exhibit 5) gives a slope of 0.61, a t-stat of 23.5 and an R-squared of 0.086. Over 1,000 pairs, 96.6% (SE 0.6%) have $|\text{t-stat}| > 1.96$; the median $|\text{t-stat}|$ is 36 (batch SE 0.9) and the median R-squared 0.18 (batch SE 0.007). On one year of 246 sessions (experiment 4) the share is still 81.7%. Regressing Δy_t on Δx_t for the first pair gives a t-stat of −0.98.

Exhibit 5: Two independent random walks



Note: the first pair of experiment 2 (seed `[2, 20261011]`), 5,850 sessions of iid standard normal steps; the walks end at 62.0 and 98.3. Simulated, not market data. *Model.*

The classical standard error (23.2) assumes white-noise errors, but here the residual is itself close to a random walk. Such a fit is a **spurious regression** (Granger and Newbold 1974; Greene 2018, ch. 21). Differencing restores a valid test, but it discards the levels.

DEFINITION **Cointegration**

Two series x_t and y_t, each with a unit root, are **cointegrated** when some combination $y_t - \beta x_t$ is stationary. The vector $(1, -\beta)$ is their **cointegrating vector**.

The log basis is such a combination with the known vector $(1, -1)$, so a unit-root test on b_t is a cointegration test, and Example 6 rejects a unit root. Whether futures and spot prices are cointegrated depends on the time-series properties of the cost of carry (Brenner and Kroner 1995). With an estimated vector, the **Engle–Granger** test applies a unit-root test to the OLS residual, with its own critical values (Engle and Granger 1987; MacKinnon 2010).

In an **error-correction model** each series' change responds to yesterday's gap b_{t-1} through an **adjustment coefficient**, α_F for the future and α_S for the index (Engle and Granger 1987).

EXAMPLE 8

Engle–Granger and error correction for TX and TAIEX

Regress $\ln F_t$ on a constant and $\ln S_t$ over TX23 and test the residual. Then regress the roll-free $\Delta \ln F_t$ and $\Delta \ln S_t$ on a constant and b_{t-1}, with White (HC0) standard errors.

SOLUTION

OLS gives $\ln F_t = -0.0283 + 1.0028 \ln S_t + \hat{\varepsilon}_t$. statsmodels' `coint` gives −9.42 against its 5% point −3.34 (one-sided), a p-value below 10^{-14}: it rejects the null of no cointegration. The residual's lag-1 autocorrelation is 0.764, so the slope's classical SE of 0.000154 and its "t-stat" of 18 against $\beta = 1$ mean nothing. Theory fixes the vector at $(1, -1)$; $\hat{\beta} > 1$ fits a basis whose mean drifts up with the index level (Example 6).

The adjustment coefficients are $\hat{\alpha}_F = -0.1597$ (SE 0.055) and $\hat{\alpha}_S = +0.0655$ (SE 0.040). A high basis predicts a fall in F and, less clearly, a rise in S: the future does most of the correcting. Their difference, −0.2252, is the $\hat{\delta}$ of the DF regression on roll-free changes.

With more than two series, Johansen's (1988, 1991) test counts the cointegrating vectors. Threshold models let the basis revert only outside a band (Dwyer, Locke and Yu 1996; Lin, Chen and Hwang 2003, for Taiwan). Pairs trading is not in this book.

PITFALL **Testing $\hat{\beta} = 1$ with the classical standard error**

In a cointegrating regression the regressor has a unit root and the residual is autocorrelated, so OLS's classical standard error is meaningless. When theory fixes the vector, test b_t for a unit root instead.

導讀 **虛假迴歸與共整合**

兩條互相獨立（independent）的隨機漫步（random walk）做迴歸（regression），t 統計量（t-statistic）卻常常大得驚人：一千組模擬中有 96.6% 的 $|\text{t-stat}|$ 超過 1.96，這就是虛假迴歸（spurious regression）。原因是殘差（residual）本身也像隨機漫步，古典標準誤（classical standard error）假設的白噪音（white noise）根本不成立。共整合（cointegration）是例外：兩條都有單根（unit root），但某個線性組合（linear combination）是平穩（stationary）的。期貨與現貨的對數基差（log basis）就是向量（vector）$(1, -1)$ 的組合，所以對基差做單根檢定（unit root test），就是在檢定共整合。誤差修正模型（error correction model）描述兩者如何把昨天的缺口拉回來，在台指期上，主要是期貨在修正。

KNOWLEDGE CHECK 5

The first pair of Exhibit 5 gives a t-stat of 23.5 in levels and −0.98 in differences. Why not difference $\ln F_t$ and $\ln S_t$ as well?

ANSWER

Differencing removes the spurious fit, but on $\ln F_t$ and $\ln S_t$ it would also remove the long-run link that the error-correction term b_{t-1} keeps.

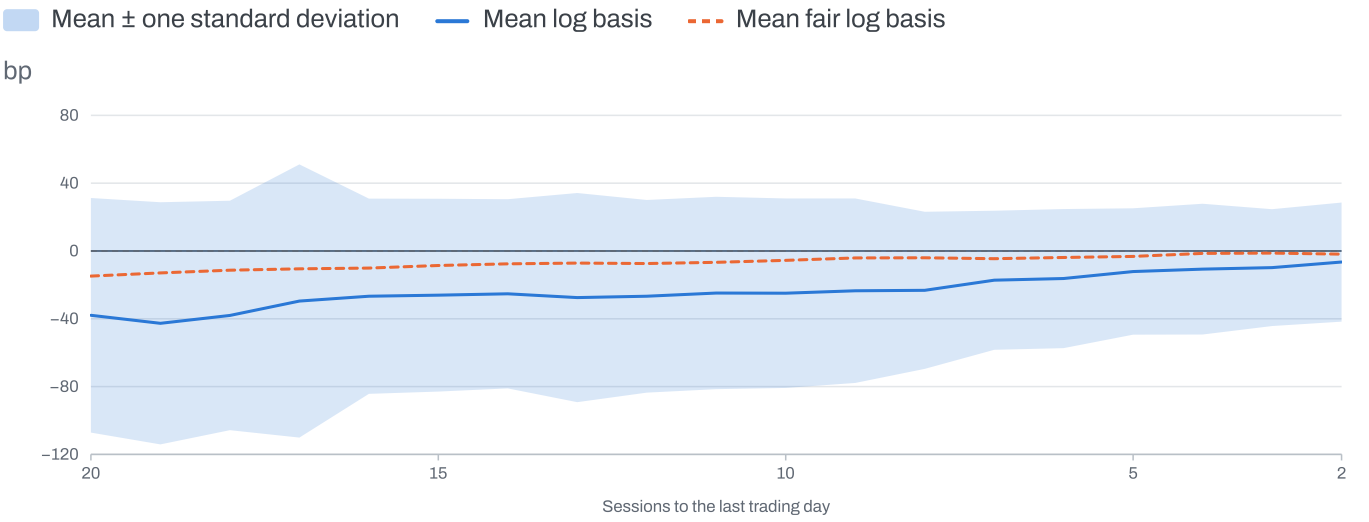
7 THE TX BASIS AS A TIME SERIES

- ☐ decompose the TX–TAIEX log basis into convergence at expiry, roll jumps, the dividend season and one-day noise, and measure how fast it reverts

In logs, LM6's (6.22) reads $b_t = (r_f - q)\tau$. Take the dividends that actually went ex, $q_{\text{real}}\tau$ from the total-return index (LM6). Then the **fair log basis** is $r_f\tau - q_{\text{real}}\tau$, and the basis minus it is the residual of (6.25) in log form. The basis has three slow parts and one fast one.

Convergence at expiry. Carry and dividends shrink with τ (Exhibit 6). Twenty sessions before the last trading day the basis averaged −37.95 bp (standard deviation 69.17) against a fair −14.78; two sessions before, −6.56 bp (35.13) against −1.84. The standard deviation falls only to about 35 bp, of which one-day noise (22.2 bp, Example 10) is a part.

Exhibit 6: The log basis as the last trading day nears

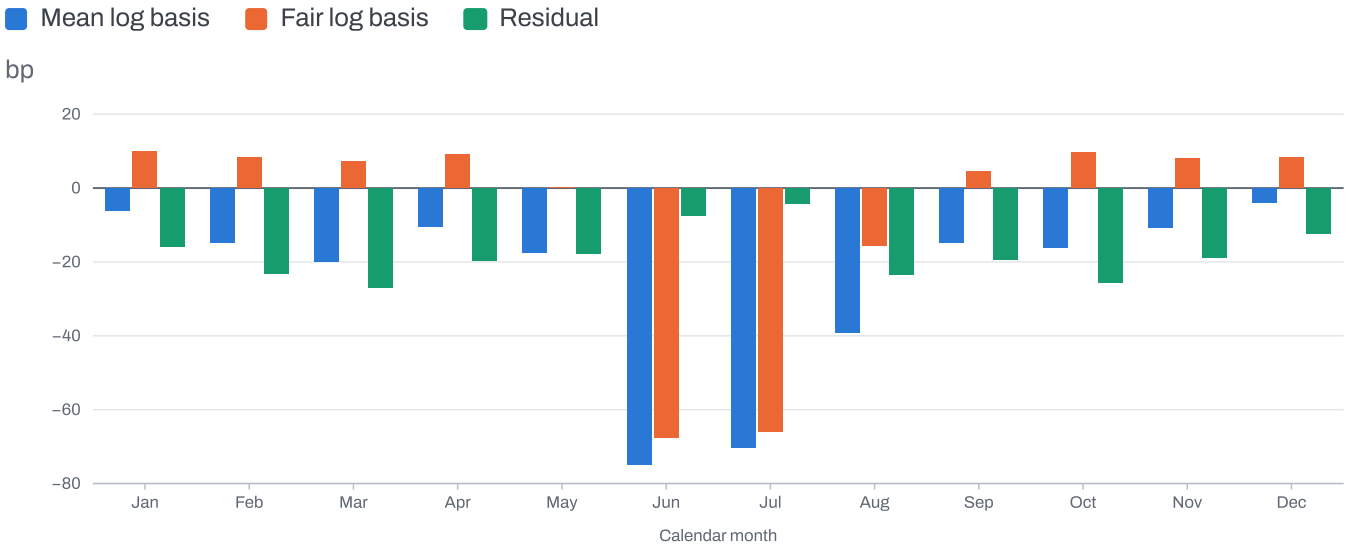


Note: sample D, 235 to 285 dates per point. Descriptive. Source: FinMind `TaiwanFuturesDaily` (TX), `TaiwanStockPrice` and `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 11 to 19 October 2026.

Roll jumps. On a roll date the series switches to a contract with a longer life. The 285 roll jumps, rolled minus roll-free change, averaged −37.5 bp (standard deviation 58.7).

The dividend season. Exhibit 7 splits the calendar-month means. June's −74.80 bp is −67.43 fair plus −7.37 residual, and July's −70.17 is −65.88 plus −4.29. Over D the basis averaged −25.57 bp: carry +9.73, dividends −17.54 and residual −17.76.

Exhibit 7: Mean log basis by calendar month, with its fair part and residual



Note: sample D, 5,834 dates, 369 to 532 a month; bp. Dividends are counted on their ex-dates up to each contract's last trading day. Means only, without standard errors: neighbouring days correlate 0.775, and the means describe the sample. Descriptive. Source: as Exhibit 6.

EXAMPLE 9

Four versions of the basis

On D, test four series for a unit root and fit AR(1) and ARMA(1,1) models: b_t; $b_t - r_f\tau$; the residual $b_t - r_f\tau + q_{\text{real}}\tau$; and b_t minus its calendar-month mean.

SOLUTION

Every version rejects a unit root (DF −27.2 to −32.3, ADF −7.75 to −9.15) and fails KPSS at the 5% level (Exhibit 8). The AR(1) half-lives run from 1.92 to 2.72 sessions and the ARMA ones from 5.05 to 7.01. Removing slow parts shortens the half-lives by up to 29%, but every version still reverts within days.

Exhibit 8: Four versions of the log basis: unit-root tests and fitted models

Version	Mean (bp)	s (bp)	DF	ADF (lags)	KPSS (p-value)	AR(1) $\hat{\phi}_1$ (half-life)	ARMA $\hat{\phi}_1, \hat{\theta}_1$ (half-life)
b_t	−25.6	58.11	−27.2	−9.03 (28)	0.62 (0.021)	0.7751 (2.72)	0.90584, −0.355 (7.01)
$b_t - r_f\tau$	−35.3	59.11	−27.4	−9.15 (28)	0.62 (0.021)	0.7723 (2.68)	0.89557, −0.327 (6.28)
Residual	−17.8	47.67	−32.3	−7.75 (33)	0.77 (<0.01)	0.6974 (1.92)	0.89075, −0.414 (5.99)
b_t − month mean	0.0	53.24	−30.3	−8.51 (28)	0.96 (<0.01)	0.7284 (2.19)	0.87168, −0.323 (5.05)

Note: sample D, so the first row differs slightly from Example 6. DF with a constant and no lags; ADF with a constant, lags by AIC. The 5% points are about −2.86 (MacKinnon) and 0.463 (KPSS); KPSS p-values stop at the table's bound, 0.01. Half-lives by (25.2) from the printed coefficients; ARMA fits at the likelihood maximum. In-sample. Source: as Exhibit 6.

Now suppose the basis is a persistent AR(1) plus one-day noise: $b_t = x_t + u_t$, with $x_t = c + \phi_1 x_{t-1} + \zeta_t$ and ζ_t, u_t independent white noise. The quasi-difference

$$b_t - \phi_1 b_{t-1} = c + \zeta_t + u_t - \phi_1 u_{t-1}$$

is correlated with its own lag only. An MA(1) $\eta_t + \theta_1 \eta_{t-1}$ has the same autocovariances when

$$\begin{aligned} \sigma_u^2 &= -\frac{\theta_1 \sigma_\eta^2}{\phi_1}, \\ \sigma_\zeta^2 &= (1+\theta_1^2)\,\sigma_\eta^2 - (1+\phi_1^2)\,\sigma_u^2 \end{aligned} \tag{25.9}$$

(Problem 8). So b_t has ARMA(1,1) autocorrelations with $\theta_1 < 0$, and a fitted ARMA(1,1) splits it into the two parts. statsmodels fits it by maximum likelihood (LM20), with standard errors from the curvature $-\ell''$, (20.6). Their large-sample law for a stationary ARMA that is invertible ($|\theta_1| < 1$) is stated without proof.

EXAMPLE 10

A persistent basis plus one-day noise

Fit an ARMA(1,1) to b_t (bp, TX23) at the likelihood maximum and read it with (25.9). Give the persistent part's half-life with a delta-method interval.

SOLUTION

statsmodels' default fit stops 0.006 below the maximum log-likelihood, so the fit starts from `method="innovations_mle"`, and a restart does not raise ℓ. At the maximum the mean is −25.5 bp (SE 3.2), $\hat{\phi}_1 = 0.90620$ (SE 0.0071), $\hat{\theta}_1 = -0.3553$ (SE 0.017) and $\hat{\sigma}_\eta^2 = 1{,}257.0$ bp².

By (25.9), $\hat{\sigma}_u = 22.2$ bp and $\hat{\sigma}_\zeta = 22.8$ bp, and the persistent part has standard deviation $\hat{\sigma}_\zeta / \sqrt{1 - \hat{\phi}_1^2} = 53.8$ bp. The noise is 64% of the variance of a day's change and 14.5% of the level's. The model's daily-change standard deviation, 39.11 bp, matches the sample's 38.91.

The persistent part's half-life is $\ln \frac{1}{2} / \ln 0.90620 = 7.04$ sessions. The delta method (20.8) with $\mathrm{SE}(\hat{\phi}_1) = 0.007055$ gives a standard error of 0.556 and a 95% interval of 5.95 to 8.13 sessions.

The standard errors of Example 10 assume that the model is right, yet the basis's volatility changes across years (Example 1). The likelihood's sandwich, the analogue of (23.6) (stated without proof), gives $\mathrm{SE}(\hat{\phi}_1) = 0.01462$ and an interval of 4.78 to 9.30 sessions. LM26 meets it as the robust standard errors of a GARCH fit.

Stability check: before and from 1 June 2015, $\hat{\phi}_1$ is 0.90 and 0.91 and $\hat{\sigma}_u$ 25.74 and 17.45 bp (Problem 8); the DF t-stats are −20.20 and −17.88. In 2025 alone $\hat{\sigma}_u = 23.4$ bp, about 56 index points at the year's mean close of 23,919.

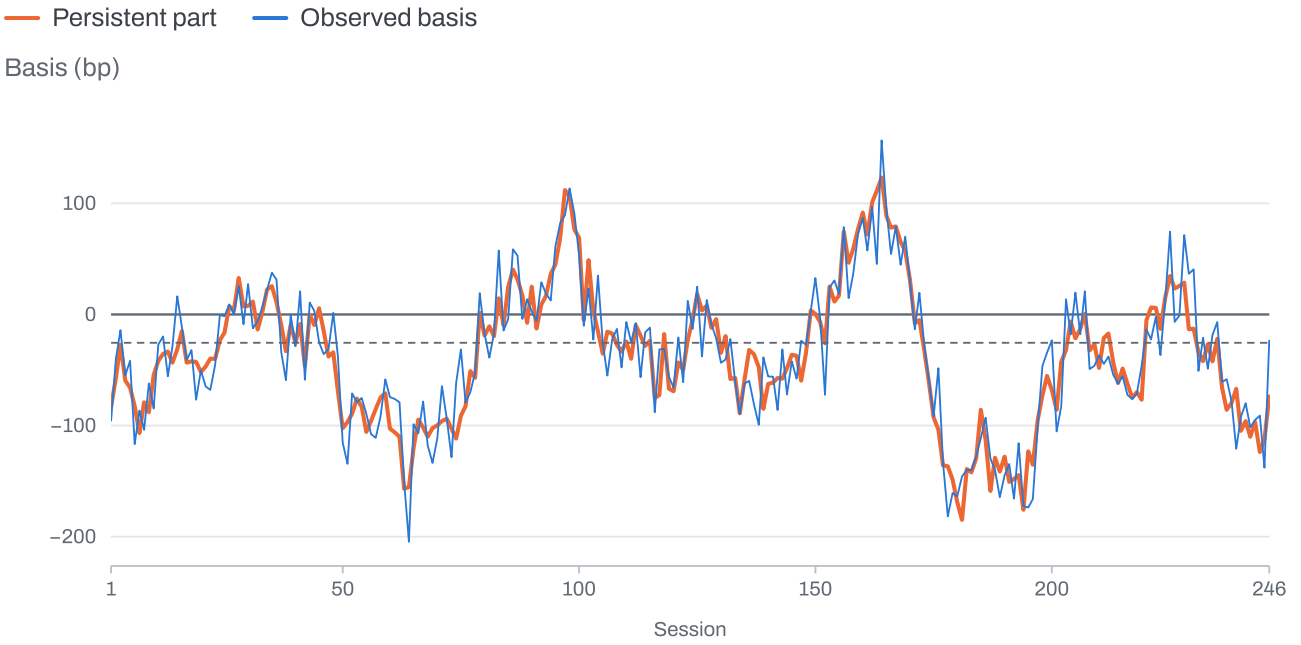
Measured in points, the 242 rolled daily changes within 2025 had a standard deviation of 81.4 (divisor 242), and the roll-free ones 76.7. LM6's 83.4 (divisor 137) used same-contract pairs from April to October only, with another roll rule.

At the 5% level KPSS rejects every version, consistent with a mean that moves across years. A fixed $r_f = 2\%$ also leaves 23 years of rate changes in the residual, which FinMind's data cannot measure. One candidate source of the noise is timing: TAIEX closes on a 13:25–13:30 call auction, while TX settles from the last minute before 13:45 (LM6). This book does not measure that gap. In continuous time a mean-reverting spread is an Ornstein–Uhlenbeck process (LM39).

The widget simulates the model over 246 sessions, with $\sigma_\zeta = 22.8$ bp around −25.5 bp. Each path's DF t-stat is compared with MacKinnon's 5% point for 245 changes, −2.87. At the initial $\phi_1 = 0.91$ the half-life is 7.35 sessions.

A mean-reverting basis with one-day noise

INTERACTIVE



Model half-life of the persistent part

7.35 sessions

AR(1) fitted to the observed path

0.794 · half-life 3.01 sessions

DF t-stat against −2.87

−5.30 · rejects a unit root

Path (seed)

1

Model, not market data: a persistent AR(1) with innovations of 22.8 bp around a mean of −25.5 bp (dashed), plus independent one-day noise, over 246 sessions. Normal draws from the book's seeded generator (seed = path number); a new path changes the draws, the sliders change only the model. The DF t-stat uses a constant and no lags, and its 5% point is MacKinnon's for 245 changes, −2.87.

PITFALL **Annualizing or ranking a one-day basis**

In Example 10's model, 64% of the variance of a day's change is noise. Annualizing a one-day basis, or ranking days by it, magnifies that noise (LM6). Average over days, or model the persistent part.

導讀 **基差：慢的三塊加上快的雜訊**

對數基差（log basis）裡有三個慢慢變的部分：到期前逐漸收斂、轉倉（roll）時跳到存續期較長的合約、每年夏天的除息旺季；6 月的平均 −74.80 bp 裡有 −67.43 bp 是持有成本（carry）與股利。剩下的是快的部分。把基差看成「持續的 AR(1)」加上「一日白噪音（white noise）」，ARMA(1,1) 的估計顯示一天變動的變異數（variance）有 64% 來自雜訊（noise），持續部分的半衰期（half-life）約 7 個交易日，而不是直接配 AR(1) 得到的 2.7 個交易日。基差幾天內就回到平均數（mean），所以單根檢定（unit root test）拒絕單根；但這個平均數會隨年份移動，所以 KPSS 也拒絕平穩（stationary）的虛無假設（null hypothesis）。13:30 收盤與 13:45 結算的時間差，可能是雜訊的來源之一。

KNOWLEDGE CHECK 6

Rolls are grouped by the calendar month of the roll date, so the June roll moves from the June into the July contract. It lowers the near-month basis by 144.7 bp on average, the July roll by 104.9 bp and the other rolls by 19.8 bp. Why?

ANSWER

The contract entered at the June roll lives through the heart of the ex-dividend season. Over 2003–2025, on average 45% of a year's dividends went ex in its life, and 39% in the July roll's contract. So its fair basis lies far below the expiring contract's, which is near zero. The other rolls add a month's carry, but the residual discount widens with time to expiry (Exhibit 6) and outweighs it.

8 TIME SERIES IN PYTHON

- [ ] run these tools in Python with statsmodels on NumPy arrays, matching each package result to a calculation by hand

Here `b` is TX23's log basis, `F` and `S` its prices and `r` TR23's log returns, all NumPy arrays; a DataFrame output ends in `.to_numpy()`.

PYTHON

```python
import numpy as np
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import acf, adfuller, coint, kpss, pacf

rho = acf(r, nlags=30, fft=False)[1:]        # (19.5), divisor T
pac = pacf(b, nlags=20, method="ols")[1:]    # the default is "ywadjusted"
lb = acorr_ljungbox(r, lags=[10]).to_numpy() # [[Q_LB(10), p-value]]
a = adfuller(b, regression="c", autolag="AIC", result_object=True)
a0 = adfuller(b, maxlag=0, autolag=None, result_object=True)
k = kpss(b, regression="c", nlags="auto", result_object=True)
eg = coint(np.log(F), np.log(S), trend="c", return_results=True)
model = ARIMA(1e4 * b, order=(1, 0, 1))      # basis in bp; const = mean
f0 = model.fit(method="innovations_mle")
f = model.fit(start_params=f0.params, cov_type="approx")  # restart, -l''
print(a.statistic, a.lags, a0.statistic, f.params, f.bse)
```

With `result_object=True`, `adfuller` and `kpss` return named fields instead of the old tuple, which now comes with a FutureWarning; `coint` takes `return_results` instead. The default `cov_type` of an ARIMA fit, `"opg"`, gives $\mathrm{SE}(\hat{\phi}_1) = 0.00385$, and Harvey's `"oim"` gives 0.0068, against 0.0071 from $-\ell''$: name the method.

IN PYTHON **Companion lab**

The Polars-first notebook `LM25_lab.ipynb` builds TX23 from the TX daily file with the roll rule. It rebuilds the module's exhibits and computational examples, runs the four simulations and solves the computational practice problems. It ends with the basis's half-lives year by year.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

`adfuller(b)` reports −9.00 with 28 lags. How do you reproduce the hand DF t-stat of −27.20?

ANSWER

Pass `maxlag=0, autolag=None`, which fits (25.7) with a constant and no lagged changes.

SUMMARY

- A random walk's variance grows like t, so a price level has no mean to return to; white noise needs zero correlation, not independence.
- An AR(1) has $\rho_k = \phi_1^k$ and half-life $\ln \tfrac{1}{2} / \ln \phi_1$, (25.1)–(25.2); an MA(1) has $\rho_k = 0$ beyond lag 1, (25.3).
- Clustering widens the band of $\hat{\rho}_k$, (25.4); Ljung–Box (25.5) and the iid variance-ratio test (25.6) need iid data.
- Under a unit root without drift, the DF t-stat (25.7) is shifted left: use MacKinnon's −2.86, not −1.645. ADF (25.8) adds lagged changes; KPSS reverses the null.
- Independent random walks give spurious regressions; the log basis is a cointegrating combination.
- As an AR(1) plus noise, (25.9), the basis puts 64% of the variance of a day's change in noise, and its persistent part halves in about 7 sessions.
- statsmodels matches each hand calculation; an ARIMA fit's standard errors depend on its `cov_type`.

PRACTICE PROBLEMS

1. An AR(1) has $\phi_1 = 0.8$. Its autocorrelation at lag 3 is *closest* to:

   A. 0.512

   B. 0.8

   C. 0

2. An AR(1) of daily data has $\phi_1 = 0.9$. Its half-life is *closest* to:

   A. 5.0 sessions

   B. 6.58 sessions

   C. 10 sessions

3. An MA(1) has $\theta_1 = 0.5$. Its ρ_1 is *closest* to:

   A. 0.5

   B. 0.25

   C. 0.40

4. A DF regression with a constant on 5,849 changes gives a t-stat of −2.50. At the 5% level (one-sided, H_0: a unit root without drift), the test:

   A. rejects, since −2.50 is below −1.645.

   B. does not reject, since MacKinnon's 5% point is −2.86.

   C. rejects, since $|\text{t-stat}| > 1.96$.

5. On TX23 the basis's ADF test rejects a unit root, and KPSS rejects stationarity around a constant (p-value 0.017). The best reading is:

   A. the basis returns within days to a mean that itself moves across years.

   B. the basis has a unit root, because KPSS rejects.

   C. the tests contradict each other, so neither informs.

6. TR23's log returns ($T = 5{,}849$) have $\hat\rho_1, \dots, \hat\rho_5$ = 0.0393, 0.0000, −0.0031, −0.0323, −0.0173, with robust standard errors (25.4) of 0.0205, 0.0210, 0.0227, 0.0178 and 0.0196. Compute $Q_{\mathrm{LB}}(5)$ and compare it with the χ^2_5 5% point 11.07. Compute the five robust t-stats (two-sided, 5% level). Why does Ljung–Box reject while no robust t-stat does? With five tests, what does the count imply?

7. From weak stationarity and (25.1), derive $\mathbb{E}x_t = c/(1-\phi_1)$, $\operatorname{Var} x_t = \sigma_\eta^2/(1-\phi_1^2)$ and $\rho_k = \phi_1^k$, and then the half-life (25.2).

8. (a) Derive (25.9) from $b_t = x_t + u_t$. (b) Apply it to the ARMA(1,1) fits of the two limit regimes, printed to four decimals: $\hat\phi_1 = 0.9035$, $\hat\theta_1 = -0.3606$, $\hat\sigma_\eta^2 = 1{,}659.9$ bp² before 1 June 2015 and 0.9101, −0.3427, 808.6 bp² from then. Give $\hat\sigma_u$, the persistent standard deviation and the noise share of the variance of a day's change, and say what changed between the regimes.

9. *(Python)* On TX23, run the DF regression (25.7) on b_t by hand (rolled changes, constant, no lags, classical SE) and check it against `adfuller(b, maxlag=0, autolag=None)`. Repeat with roll-free changes. Give both t-stats and both half-lives, to two decimals.

10. *(Python)* Experiment 3 (seed `[3, 20261011]`) draws 100,000 random walks of 246 levels, in chunks of 500. Find the 5% point of the DF t-stat (constant, no lags) with LM18's batch SE (20 batches), and MacKinnon's 5% point for 245 observations. Find also the share below −1.645, with its binomial SE. Quantiles use `inverted_cdf`; give the points to two decimals and the share in % to one.

11. *(Python)* Experiment 4 (seed `[4, 20261011]`) draws 1,000 independent pairs of random walks of 246 sessions. Each pair is one (2, 246) array of standard normal steps, x_t then y_t, as in experiment 2. Regress y_t on a constant and x_t with the classical SE. Find the share with $|\text{t-stat}| > 1.96$ (binomial SE) and the median R-squared (`inverted_cdf`) with LM18's batch SE, in % to one decimal and to two decimals.

12. *(Python)* On TR23's log returns, compute LM19's $\mathrm{VR}(5)$ and its iid and robust t-stats (25.6), two-sided at the 5% level, using the robust standard errors of Problem 6.

SOLUTIONS

1. **A** is correct: $\rho_3 = 0.8^3 = 0.512$ by (25.1). B is ρ_1, as if an AR(1) correlated only one lag back. C is the partial autocorrelation at lag 3.

2. **B** is correct: $\ln\frac{1}{2}/\ln 0.9 = 6.58$ sessions by (25.2). A is $0.5/(1-\phi_1)$, a linear decay. C is $1/(1-\phi_1) = 10$, the sum of the weights ϕ_1^k, not a half-life.

3. **C** is correct: $0.5/(1+0.25) = 0.40$ by (25.3). A is θ_1 itself, and B is θ_1^2.

4. **B** is correct: under a unit root without drift the DF t-stat is shifted left, and −2.50 lies above −2.86. A uses the normal one-sided point. C uses a two-sided normal test.

5. **A** is correct: the two tests have different nulls, and a mean that moves across years fails KPSS while the reversion within days rejects a unit root. B forgets that KPSS's alternative includes a moving mean. C misses that both nulls can be false.

6. By (25.5),

$$\begin{aligned} Q_{\mathrm{LB}}(5) &= 5849 \times 5851 \times \sum_{k=1}^{5} \frac{\hat{\rho}_k^2}{5849-k} \\ &= 16.96 \end{aligned}$$

   from the printed values (16.98 unrounded), above 11.07, so Ljung–Box rejects. The robust t-stats are 1.92, 0.00, −0.14, −1.81 and −0.88, none beyond 1.96. Ljung–Box gives each $\hat{\rho}_k$ the iid variance $1/T$, but clustering makes the robust standard errors 0.0178 to 0.0227, against 0.013076. So Q_{LB} is too large. Five tests at the 5% level expect 0.25 false rejections, and none occurs.

7. Weak stationarity fixes $\mathbb{E}x_t$, so $\mathbb{E}x_t = c + \phi_1 \mathbb{E}x_t$ gives $c/(1-\phi_1)$, which needs $\phi_1 \neq 1$. Since η_t is uncorrelated with x_{t-1}, $\gamma_0 = \phi_1^2\gamma_0 + \sigma_\eta^2$, so $\gamma_0 = \sigma_\eta^2/(1-\phi_1^2)$, positive only if $|\phi_1| < 1$. Multiplying

$$x_t - \mathbb{E}x_t = \phi_1(x_{t-1} - \mathbb{E}x_t) + \eta_t$$

   by $x_{t-k} - \mathbb{E}x_t$ and taking expectations gives $\gamma_k = \phi_1\gamma_{k-1}$, so $\rho_k = \phi_1^k$. The expected gap shrinks by ϕ_1^k in k sessions, and $\phi_1^k = \frac{1}{2}$ at $k = \ln\frac{1}{2}/\ln\phi_1$.

8. (a) The quasi-difference $b_t - \phi_1 b_{t-1}$, written out in Section 7, has variance $\sigma_\zeta^2 + (1+\phi_1^2)\sigma_u^2$ and lag-1 autocovariance $-\phi_1\sigma_u^2$ by (16.3) and (16.4), and none beyond. An MA(1) has $(1+\theta_1^2)\sigma_\eta^2$ and $\theta_1\sigma_\eta^2$. Equating gives (25.9). (b) Before 1 June 2015: $\hat{\sigma}_u = 25.74$ bp, persistent standard deviation 60.50 bp, noise share 65.2%. From then: 17.45 bp, 44.95 bp and 62.6%. The noise fell by about a third and the persistent spread by about a quarter, while $\hat{\phi}_1$ and the noise share barely moved.

9. By hand, Δb_t on a constant and b_{t-1} gives the DF t-stat −27.20, equal to `adfuller(b, maxlag=0, autolag=None)`, and $\hat{\phi}_1 = 0.7758$, a half-life of 2.73 sessions. With roll-free changes the t-stat is −29.66 and $\hat{\phi}_1 = 0.7748$, a half-life of 2.72. The roll jumps barely matter here.

10. The 5% point is −2.87 (batch SE 0.006), and MacKinnon's for 245 observations is also −2.87. The share below −1.645 is 45.7% (SE 0.16%): one year of data does not rescue the normal point.

11. The share is 81.7% (SE 1.2%), and the median R-squared is 0.17 (batch SE 0.011). A short sample gives spurious fits too.

12. The ratio is $\mathrm{VR}(5) = 1.0465$. The iid standard error $\sqrt{72/(15 \times 5849)} = 0.02865$ gives a t-stat of 1.62. The robust form with Problem 6's standard errors gives $\mathrm{SE}_{\mathrm{rob}} = 0.04573$ and a t-stat of 1.02. Neither rejects at the 5% level.

GLOSSARY

Term	中文	Meaning
Time series	時間序列	Observations of one quantity at equal time steps.
Weak stationarity	平穩	Constant mean and variance, and autocovariances that depend only on the lag.
White noise	白噪音	Mean 0, constant variance, zero correlation between dates.
Random walk	隨機漫步	$x_t = x_{t-1} + \eta_t$ (with drift, $+c$); its variance grows like t.
Autocorrelation function (ACF)	自我相關函數	ρ_k as a function of the lag k.
Partial autocorrelation function (PACF)	偏自我相關函數	The last coefficient of an OLS regression on lags 1 to k.
Autoregressive model, AR(1)	自我迴歸模型	$x_t = c + \phi_1 x_{t-1} + \eta_t$, (25.1).
Moving-average model, MA(1)	移動平均模型	$x_t = c + \eta_t + \theta_1 \eta_{t-1}$, (25.3).
ARMA(1,1)	自我迴歸移動平均模型	An AR(1) with an MA(1) error.
Half-life	半衰期	$\ln \frac{1}{2} / \ln \phi_1$ sessions, (25.2).
Robust standard error	穩健標準誤	(25.4): valid under clustering with finite fourth moments, for no autocorrelation.
Ljung–Box test	Ljung–Box 檢定	(25.5): a joint test of lags 1 to K, approximately χ^2_K for iid data.
Variance-ratio test	變異數比檢定	(25.6): $\mathrm{VR}(m) - 1$ over its standard error.
Unit root	單根	$\phi_1 = 1$ in an AR(1), a random walk.
Augmented Dickey–Fuller test	增廣 Dickey-Fuller 檢定	(25.8): (25.7) with lagged changes; one-sided, MacKinnon's points.
KPSS test	KPSS 檢定	A test of stationarity against a unit root; large values reject.
Spurious regression	虛假迴歸	A significant-looking regression between unrelated nonstationary series.

Term	中文	Meaning
Cointegration	共整合	A stationary combination of series that each have a unit root.
Error-correction model	誤差修正模型	Changes that respond to yesterday's gap from the cointegrating relation.

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